

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\gamma^{\mu\rho\nu}\nabla_\rho\Psi_\nu^A-\gamma^{\mu\nu}\gamma^{\hat{5}}\partial_{\hat{5}}\Psi_\nu^A+k\epsilon\gamma^{\mu\nu}\gamma^{\hat{5}}\Psi_\nu^A-\sqrt{2}\left(g_RR\epsilon(\sigma_3)_B^A+g_SS(\sigma_1)_B^A\right)\gamma^{\mu\nu}\Psi_\nu^B=0\;,$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\beta(k,p)=\frac{-i}{2\pi}\delta(p_2+k_2)\delta(p_3+k_3)\int dx_1\frac{e^{ik_1x_1}}{\sqrt{4\omega_p\omega_k}}\left\{i\omega_kf_p^{+*}(t'(0,x))e^{ip_1x_1'(0,x)}-\partial_t\left\{f_p^{+*}(t'(t,x))e^{ip_1x_1'(t,x)}\right\}|_{t=0}\right\}$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$W_{m(2)}=-\frac{1}{2}E_{\gamma ab}^*f_{\alpha\beta}^{\gamma}C^{\alpha a}C^{\beta b}+\bar{E}_{\gamma a}f_{\alpha\beta}^{\gamma}B^{\alpha}C^{\beta a}+\frac{1}{2}\epsilon_{ab}\bar{Q}_iR_{\alpha,j}^iR_{\beta}^j(A+Q)C^{\beta b}C^{\alpha a}.$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\Phi_{\alpha_0}^{*(1)}=N_{\alpha_1\alpha_0}n_1^{\alpha_1}+\cdots\,,\,\varphi_{\alpha_1}^{*(1)}=(-)^{\epsilon_{\alpha_0}(\epsilon(f_{\alpha_1})+1)}N_{\alpha_1\alpha_0}\nu_1^{\alpha_0}+\cdots\,,$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\left.\mathrm{Tr}K_{\delta}^{(0)}(s)\right|_{\mathcal{C}_{\beta}}=\frac{\beta}{2\pi}\left.\mathrm{Tr}K^{(0)}(s)\right|_{R^2}+\frac{1}{8\pi i s}\int_0^\infty r dr\int_A dz\frac{\exp i\frac{(\delta-\pi)}{\beta}z}{\sin\frac{\pi}{\beta}z}\exp\frac{r^2\sin^2\frac{z}{2}}{s}\quad,$$